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High-throughput computational framework for high-order anharmonic thermal transport in cubic and tetragonal crystals

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Accurate first-principles prediction of lattice thermal conductivity (κ_L) remains challenging in identifying materials with extreme thermal behavior. While the harmonic approximation with three-phonon scattering (HA + 3ph) is now routine, reliable κ_L prediction often requires higher-order anharmonic effects, including self-consistent phonon renormalization, three- and four-phonon scattering, and off-diagonal heat flux (SCPH + 3, 4ph + OD). We present a state-of-the-art high-throughput workflow that unifies these effects and apply it to 773 cubic and tetragonal crystals spanning diverse chemistries and structures. From 562 dynamically stable compounds, we assess the hierarchical impacts of higher-order anharmonicity. For around 60% of materials, HA + 3ph predictions closely match those from SCPH + 3, 4ph + OD. SCPH generally increases κ_L , by over 8 times in extreme cases, whereas four-phonon scattering universally suppresses κ_L , sometimes to 15% of the HA + 3ph value. Off-diagonal contributions are negligible in high- κ_L systems but can rival diagonal terms in highly anharmonic low- κ_L compounds. We highlight four case studies, $\text{Rb}_2\text{TlAlH}_6$, Cu_3VSe_4 , CuBr , and KTiCl_4 , that exhibit distinct extreme behaviors. This work delivers not only a robust workflow for high-fidelity κ_L dataset but also a quantitative framework to determine when higher-order effects are essential. The hierarchy of κ_L results, from the HA + 3ph to SCPH + 3, 4ph + OD level, offers a scalable, interpretable route to discovering next-generation extreme thermal materials.

Lattice thermal conductivity (κ_L) is critical for a variety of technologies. Materials with ultralow κ_L are essential for thermal insulation and thermoelectric energy conversion^{1–4}, whereas ultrahigh- κ_L materials enable efficient heat dissipation in high-power electronics^{5,6}. Such diverse applications make the discovery of materials with extreme κ_L a major research frontier.

Recently, data-driven methods have emerged as a transformative paradigm for discovering materials with extreme thermal transport properties. Machine learning (ML) surrogates trained on existing κ_L datasets (from experiments or first-principles) can rapidly explore vast chemical spaces for extreme- κ_L materials at a fraction of the computational cost^{7–9}. Deep generative models have been used to propose new crystal structures, while fast ML interatomic potentials evaluate their stability and thermal properties, accelerating the design of ultrahigh- κ_L materials^{10,11}.

These efforts are further accelerated by universal ML interatomic potentials, such as M3GNet¹², CHGNet¹³, and MACE¹⁴, now achieving near-DFT accuracy for energies and forces at orders of magnitude lower cost¹⁵.

These advances naturally create an urgent demand for high-fidelity κ_L data. Experimental data, though reliable, remain limited in both volume and chemical diversity. First-principles calculations have thus emerged as a critical source of thermal conductivity data. Recent advances in first-principles lattice dynamics have begun to accelerate the prediction of κ_L for unseen crystalline materials. High-throughput computational frameworks that automate the calculation of harmonic and anharmonic phonon properties and lattice thermal conductivity have emerged. Large phonon databases generated via the finite-displacement method (PhononDB)¹⁶ and density-functional perturbation theory¹⁷ have gained broad utility across the community. Ohnishi et al.¹⁸ developed the “auto-kappa” workflow, which

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computed κ_L for over 6000 materials under the harmonic approximation with three-phonon scattering and off-diagonal heat flux considered. Zhu et al.¹⁹ introduced an automated workflow that can extract up to the fourth-order interatomic force constants and calculate κ_L , thermal expansion, and vibrational free energies, with demonstrations on dozens of compounds.

Despite these efforts, achieving predictive accuracy for κ_L remains challenging, particularly in strongly anharmonic materials. It is now well established that both three-phonon and four-phonon scattering are important to reliably uncover microscopic mechanisms underlying thermal transport processes^{20,21}. Moreover, finite-temperature phonon renormalization²² and off-diagonal heat flux (wave-like phonon tunneling)²³ can strongly affect κ_L , especially in highly anharmonic systems. Our recent study highlights that only the most advanced lattice-dynamics treatments—combining self-consistent phonon renormalization (SCPH) with three- and four-phonon scattering (3, 4ph) and off-diagonal heat flux (OD)—can reliably predict κ_L in challenging materials²⁴.

To fully enable data-driven discovery of extreme- κ_L materials, a crucial step is to generate consistent, high-fidelity thermal conductivity datasets using state-of-the-art lattice dynamics within efficient, scalable workflows. Integrating such workflows into established materials databases (e.g., Materials Project²⁵, OQMD²⁶, and AFLOW²⁷) holds promise for building large-scale thermophysical property datasets, thereby accelerating data-driven materials discovery and the training of machine-learned interatomic potentials.

In this work, we introduce a high-throughput first-principles workflow that computes κ_L with full anharmonic corrections. Our automated pipeline generates second-, third-, and fourth-order force constants, performs self-consistent phonon renormalization, and solves the phonon Boltzmann equation, including both diagonal (Peierls)²⁸ and off-diagonal (Allen-Feldmann)²⁹ contributions. We apply this framework to a diverse dataset of 773 cubic and tetragonal crystals, spanning dozens of chemical elements and multiple structural prototypes, including elementals, binary zinc blendes and rocksalts, ternary perovskites, Heuslers, and chalcopyrites, quaternary double-perovskites, etc. For every compound, we computed a hierarchy of κ_L values: starting from the baseline of the harmonic approximation with three-phonon scattering (HA + 3ph), adding SCPH renormalization (SCPH + 3ph), including four-phonon scattering (SCPH + 3, 4ph), and finally incorporating off-diagonal transport (SCPH + 3, 4ph + OD). We also introduce hierarchical metrics to quantify the relative influence of each anharmonic effect.

From our data-driven analysis, clear trends emerge: 1) For most materials, the HA + 3ph prediction closely approximates the fully anharmonic SCPH + 3, 4ph + OD value, suggesting higher-order corrections may often be skipped due to insignificant anharmonicity or countervailing effects; 2) SCPH corrections typically increase κ_L due to phonon hardening, though rare cases show reductions; 3) Four-phonon scattering consistently reduces κ_L , with effects ranging from modest in high- κ_L compounds to pronounced in strongly anharmonic systems. Strong four-phonon effects are also found in some high- κ_L crystals, possibly linked to acoustic phonon bunching; 4) Off-diagonal contributions are significant only in low- κ_L materials, where wave-like transport dominates due to phonon linewidth broadening and band clustering. Beyond these quantitative trends, our dataset enables the identification of chemical and structural motifs associated with extreme thermal transport properties. These insights underscore how comprehensive first-principles κ_L data can illuminate the phonon physics underlying material performance and guide the design of next-generation thermal materials.

Results

High-throughput computational workflow

We present the workflow of our high-throughput lattice dynamics and thermal transport computational framework in Fig. 1a. The software and codes utilized at each stage are highlighted in bold, accompanied by brief descriptions of their respective roles. Key outputs, including harmonic and temperature-dependent phonon dispersions as well as lattice thermal

conductivities at different levels of theory, are emphasized in yellow for clarity. A step-by-step guide to each module of the workflow, including detailed parameter settings and procedural logic, is provided in the subsequent sections. Most parameters are adopted from our previously validated high-throughput studies^{24,30}. In addition, two new parameters, q-mesh density d and the number of four-phonon scattering processes sampled N_{sample} , have been rigorously tested using a diverse representative set of 33 compounds. Comprehensive convergence criteria, validation results, and a detailed breakdown of computational time costs are included in the Supplementary Information. Detailed instructions for each step are as follows:

1. **VASP relaxation and self-consistent field (SCF) calculations:** As the first step of our high-throughput workflow, we perform density functional theory (DFT) calculations to relax retrieved crystal structures from the Materials Project²⁵, using the Vienna Ab initio Simulation Package (VASP)³¹. For the exchange-correlation functional, we employ the generalized gradient approximation (GGA) of Perdew, Burke, and Ernzerhof for solids (PBEsol)³². A high plane-wave energy cutoff of 520 eV is used to ensure the accuracy of the basis set. The structural relaxation continues until the total energy and the residual forces between two consecutive self-consistent field (SCF) steps converge to within 10^{-8} eV and 10^{-3} eV Å⁻¹, respectively. To guarantee reliable forces and energies from each SCF step, particularly critical for subsequent force constant extraction, we adopt a dense k-mesh with KSPACING of 0.15 Å⁻¹ during both geometry optimization of the primitive cell and SCF calculations on displaced supercells.
2. **Small and temperature-dependent displacement generation for supercells:** Next, we use the finite displacement method (FDM), as implemented in Phonopy³³, to extract the second-order interatomic force constants (IFCs), $\Phi^{(2)}$, under the harmonic approximation, and to compute the corresponding harmonic phonon dispersions ω_λ . For accurate ω_λ in ionic compounds, non-analytical corrections (NAC) are required to capture the longitudinal optical (LO)–transverse optical (TO) splitting at the Brillouin zone center. Given the focus of this work, we are more concerned with the influence of NAC on the κ_L results. In the Supplementary Information, we perform a systematic evaluation of NAC effects on five of the most ionic compounds in our dataset. We find that the upper bound of NAC-induced variation in $\kappa_L^{\text{HA}+3\text{ph}}$ is $\sim 12\%$. This impact decreases as the theory level elevates, to $\sim 7\%$ in $\kappa_L^{\text{SCPH}+3,4\text{ph}}$. These errors can be considered as non-essential in the context of our baseline high-throughput framework, particularly because NAC primarily affects the dispersion of high-frequency optical modes, which typically contribute little to κ_L in most materials. Therefore, we opt not to include NAC in our phonon and κ_L calculations. This decision is aligned with prior findings by Simoncelli et al.³⁴, who also concluded that NAC impact on κ_L is generally minor. Nonetheless, we advise users intending to apply our workflow to highly ionic materials or systems where optical phonons play a dominant role in thermal transport to consult the Supplementary Information. There, we provide a detailed guide on how to augment NAC effects into the results generated by the current workflow. For supercell set construction, we generate diagonal-only supercells with an average number of atoms of ~ 120 , and set the minimum number of atoms in a supercell as 40 to balance computational efficiency and convergence reliability. As mentioned in the Methods section, temperature-dependent atomic displacements in supercells are generated using the quantum covariance matrix approach implemented in the TD-Disp code³⁵. The ω_λ in Eq. (17) is forced to be positive if there is imaginary mode occurring ($\omega_\lambda = |\omega_\lambda|$), which is a relatively simple treatment while preserving the physical relevance of generated displacements. A similar approach has been employed in the first iteration of solving SCPH by Tadano et al.²². To ensure statistically sufficient sampling for high-order IFC fitting, we determine the number of displaced supercells such that the total number of available force components ($3 \times N_{\text{atom}} \times N_{\text{supercell}}$) is at least five times greater

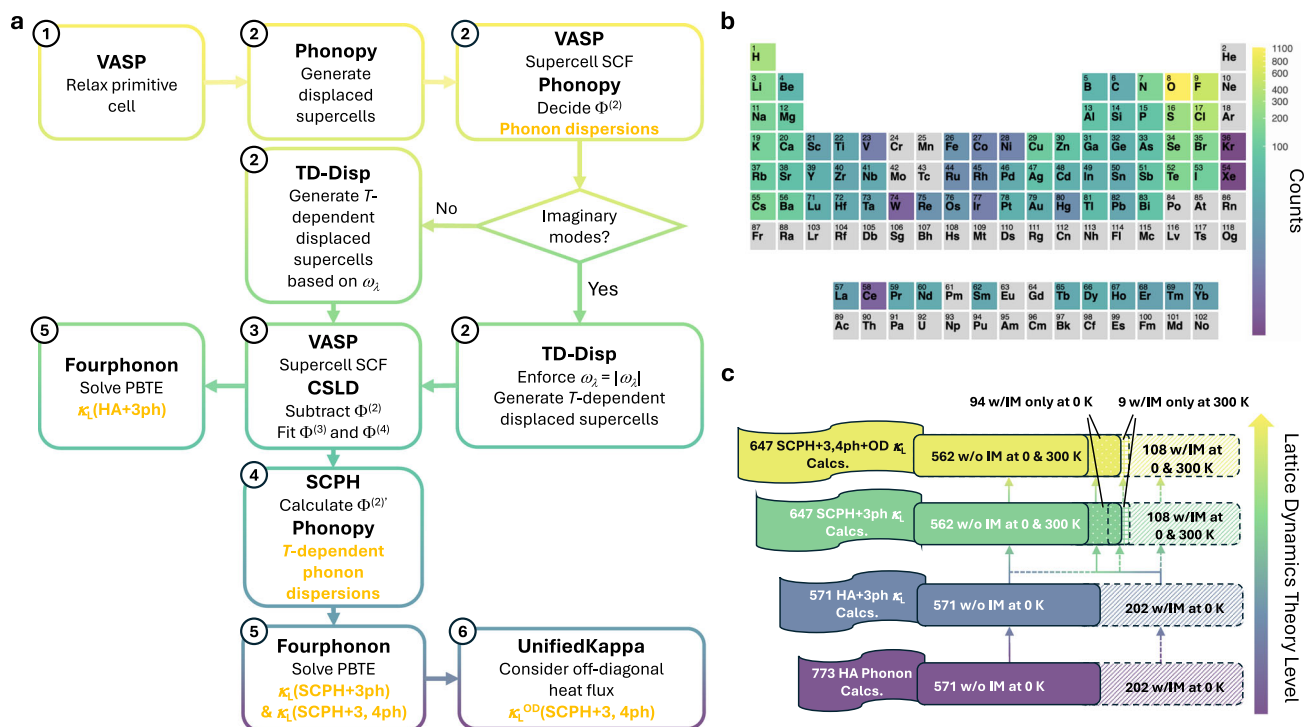


Fig. 1 | Overview of the high-throughput computational framework and dataset composition. **a** Workflow for high-throughput lattice thermal conductivity calculations. Outputs discussed in this work are highlighted in yellow. **b** Elemental distribution of 773 materials included in the dataset. **c** Dataset overview across the lattice-dynamics theory hierarchy. At each theory level, solid boxes/arrows trace

materials without imaginary modes (IM) at 0 K (under the harmonic approximation, HA) and 300 K (after the self-consistent phonon renormalization, SCPH), whereas dashed boxes/arrows show materials with IM at 0 K (HA) and 300 K (SCPH). The banner on each row reports the number of materials with data available at that level in our released dataset.

than the number of independent IFC parameters after symmetry reduction.

3. *Compressive sensing lattice dynamics (CSLD) force-constant fitting:* After completing SCF calculations on all displaced supercells, we collect the resulting atomic forces to form the training dataset for CSLD^{36,37}. Specifically, 95% of the data is used for model training and the remaining 5% is reserved for testing. To optimize the regularization parameter μ , we perform a grid search spanning four orders of magnitude, i.e. $\mu = 10^{-5}$, 10^{-7} , 10^{-9} , and 10^{-10} . We adopt the “cocktail”-flavor approach³⁸, where the harmonic $\Phi^{(2)}$ is first subtracted from the total forces, and the residual forces are then used to fit $\Phi^{(3)}$ and $\Phi^{(4)}$. When fitting the IFCs, we consider the cutoff radii of three- and four-body interactions up to the 4th and 2nd nearest neighbor, respectively. These cutoff settings are adopted based on a series of thorough convergence tests in our previous work in cubic systems^{21,24,30,39}. Due to limited computational resources, we constrain the scope of this study to cubic and tetragonal crystals. For applications to lower-symmetry structures, we strongly recommend performing careful convergence tests with respect to the cutoff radii to ensure the accuracy of IFC fitting. A trained LASSO model is accepted only if it achieves a mean absolute percentage error (MAPE) below 20% on the testing set. Models meeting this threshold are deemed reliable for extracting the final high-order IFCs. Readers seeking further details on this metric are referred to the Supplementary Information for a comprehensive discussion.
4. *Self-consistent phonon renormalization (SCPH):* With well-fitted $\Phi^{(4)}$, we proceed to perform SCPH calculations at 300 K based on our in-house implementation²⁴. The phonon frequencies are renormalized iteratively in reciprocal space using a q-mesh that is twice as dense in each dimension as the one corresponding to the supercell used in the IFC fitting procedure. Given that our SCPH formalism operates in reciprocal space, this effectively corresponds to a real-space supercell that is doubled in size relative to the one employed during the force-

constant extraction. The SCPH iterations are deemed converged when the maximum change in phonon frequencies Ω_λ between successive steps falls within 10^{-3} THz.

5. *Lattice thermal conductivity calculations:* We compute the lattice thermal conductivity κ_L at 300 K by solving the PBTE iteratively (three-phonon) or under the relaxation time approximation (four-phonon) using the FourPhonon^{40,41} code. We specially maintain the hierarchy of lattice dynamics theories by calculating κ_L respectively with i) harmonic approximation and three-phonon scattering process ($\kappa_L^{\text{HA}+3\text{ph}}$), ii) self-consistent phonon renormalization and three-phonon scattering process ($\kappa_L^{\text{SCPH}+3\text{ph}}$), and iii) self-consistent phonon renormalization and three- and four-phonon scattering process ($\kappa_L^{\text{SCPH}+3,4\text{ph}}$). To ensure uniform sampling of phonon modes across different crystal systems, we define a consistent q-mesh density d as

$$d = n_i \cdot \|\mathbf{l}_i\|, i = \alpha, \beta, \gamma \quad (1)$$

where n and $\mathbf{l}$ are the number of q-points and lattice vector along the i -th crystallographic direction, respectively. Based on the convergence test (see Supplementary Information), we set $d = 65$ for both $\kappa_L^{\text{HA}+3\text{ph}}$ and $\kappa_L^{\text{SCPH}+3\text{ph}}$ calculations, and $d = 55$ for $\kappa_L^{\text{SCPH}+3,4\text{ph}}$ calculations. To speed up the four-phonon scattering rate calculations, particularly, we employ the random sampling scheme implemented in FourPhonon⁴¹, setting the number of sampled scattering events N_{sample} as 400,000. Although this number has been verified to yield converged results in our work based on systematic benchmarking (see Supplementary Information), we recommend that users perform N_{sample} convergence test towards higher values when handling a smaller and targeted dataset, which might contain hard-to-converge systems. Note that in this work, all reported κ_L values, regardless of the level of theory, refer to the directional average computed over the three

crystallographic axes, i.e. $\kappa_L = 1/3(\kappa_L^\alpha + \kappa_L^\beta + \kappa_L^\gamma)$. We do this to facilitate a straightforward comparison of κ_L across materials without introducing complications from anisotropy.

6. **Off-diagonal lattice thermal conductivity calculation:** In the final step, we compute the off-diagonal lattice thermal conductivity, κ_L^{OD} , using the Unifiedkappa code^{42,43} under the relaxation time approximation. Unless stated otherwise, all reported values of κ_L^{OD} in this work refer to the off-diagonal lattice thermal conductivity at the SCPH + 3, 4ph level.

Overview of the dataset

Our previous work demonstrated a striking contrast in the lattice thermal conductivity and underlying phonon transport mechanisms between rocksalts and zinc blends, driven by differences in quartic anharmonicity and off-diagonal heat flux contributions²⁴. Specifically, we showed that these disparities arise from the distinct local coordination and bonding environments inherent to each prototype. Motivated by these findings, the present study substantially expands the chemical and structural diversity of our analysis to 773 cubic and tetragonal materials. These materials, selected from existing entries in PhononDB, are limited to those with fewer than 20 atoms per primitive cell, and span elemental, binary, ternary, and quaternary chemistries. Beyond the previously studied rocksalts and zinc blends, this expanded dataset includes a broad spectrum of cubic and tetragonal crystal types such as perovskites, Heuslers, chalcopyrites, and sulvanites, thereby enabling a more comprehensive understanding of phonon transport across diverse structural and bonding environments. Figure 1b maps the chemical distribution of our dataset onto a periodic table. Oxygen emerges as the most frequently occurring element, reflecting the prevalence of oxides. Other anions from the chalcogen and halogen groups also appear prominently. Hydrogen, notable for its flexible role as both cation and anion, is also among the most frequent elements. On the cationic side, the dataset primarily comprises materials with main-group elements, for which DFT-calculated energies and forces are generally more reliable. Transition metals, and to a lesser extent, rare earth elements, are included at lower frequencies. Notably, high-spin transition metals such as Cr and Mn, which are known to pose challenges for DFT due to their complex magnetic behavior, have been deliberately excluded to enhance the overall robustness and accuracy of the dataset.

It is important to note that our dataset size varies across different levels of lattice dynamics theory, as illustrated in Fig. 1c. Under the harmonic approximation, phonon spectra are successfully computed for all 773 materials. However, 202 of these exhibit imaginary phonon modes (IM), i.e., negative frequencies that signal unphysical phonon eigenvalues. These IM preclude calculations of κ_L at the HA + 3ph level, resulting in only 571 materials with valid $\kappa_L^{\text{HA}+3\text{ph}}$ values. Sometimes, IM may be artifacts of unconverged computational settings, as will be discussed in the later benchmarking section. Nonetheless, in many cases, IM, also referred to as “soft modes”, may indicate dynamic instability at 0 K, motivating further investigation involving anharmonic lattice dynamics via SCPH theory. Upon applying SCPH at 300 K, 94 out of the original 202 materials with soft modes exhibit hardening, i.e., their negative frequencies shift to positive values. This behavior can be attributed to temperature-induced stabilization of certain metastable phases. A well-known example is the octahedral tilting in perovskites, which enables the stabilization of the cubic phase at finite temperature⁴⁴. Meanwhile, there are also 9 rare cases that show IM after SCPH. We give a more detailed discussion about the abnormal SCPH softening in the following section. The materials that recover from IM upon SCPH correction are subsequently included in the κ_L calculations at both SCPH + 3ph and SCPH + 3, 4ph + OD level, generating a dataset of 647 materials with valid $\kappa_L^{\text{SCPH}+3\text{ph}}$ and $\kappa_L^{\text{SCPH}+3,4\text{ph}+\text{OD}}$ values. The remaining 108 materials with persistent IM at 300 K may potentially stabilize at higher temperatures, as SCPH could further harden their soft modes. However,

such finite-temperature stabilization lies beyond the scope of our current high-throughput investigation, which focuses exclusively on room-temperature thermal conductivity. Most importantly, the central objective of this work is to study the hierarchical impact of higher-order anharmonicity. To ensure meaningful and consistent comparisons, each material included in the analysis must possess fully converged values of $\kappa_L^{\text{HA}+3\text{ph}}$, $\kappa_L^{\text{SCPH}+3\text{ph}}$, $\kappa_L^{\text{SCPH}+3,4\text{ph}}$, and κ_L^{OD} . Based on convergence criteria detailed in the previous section, we identify a subset of 562 materials that meet these requirements. Unless otherwise noted, this final set of 562 materials forms the basis for all subsequent discussions related to κ_L in this study.

Benchmark with literature and database

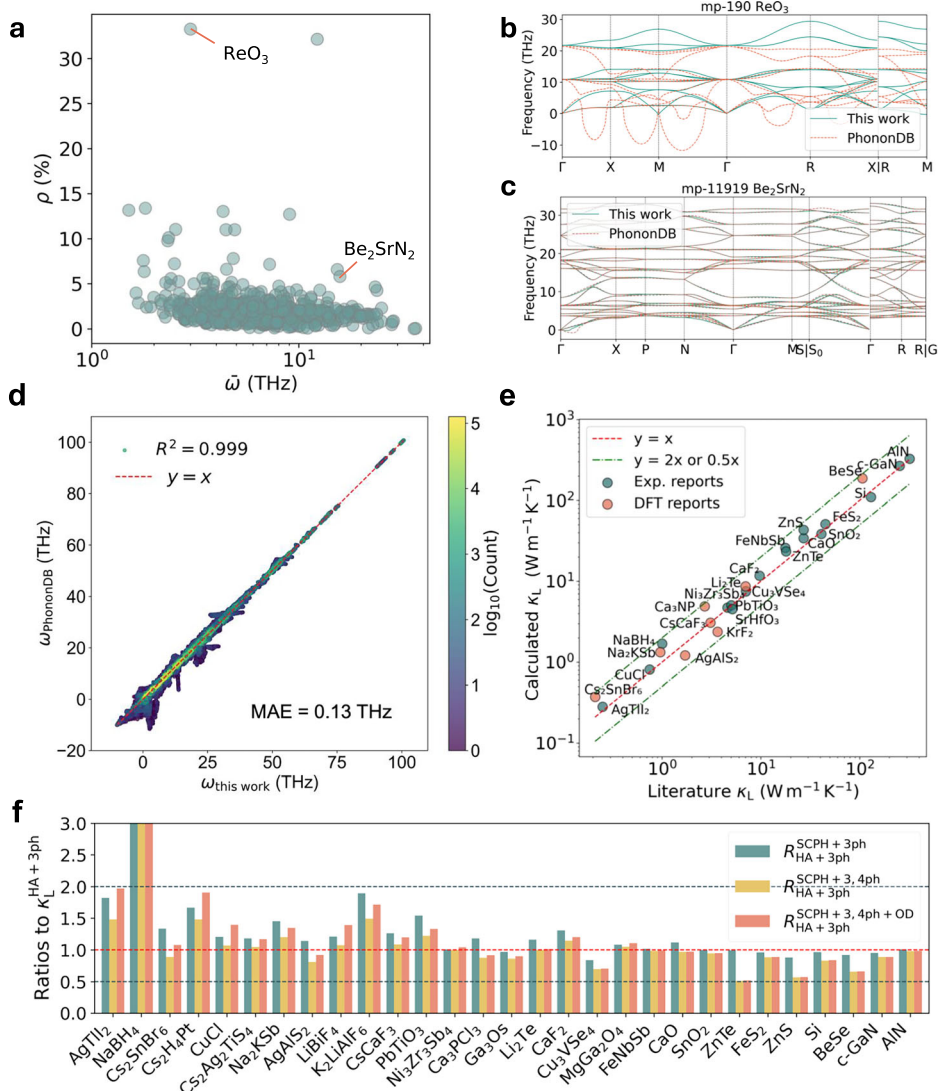
To ensure the reliability of our high-throughput workflow, we first benchmark the calculated phonon frequencies for 773 materials against the PhononDB¹⁶ results. We define the relative phonon frequency difference ρ as:

$$\rho = \frac{\|\omega^{\text{DB}} - \omega\|_2}{\|\omega^{\text{DB}}\|_2} \times 100\%, \quad (2)$$

where ω^{DB} and ω are arrays of phonon mode frequencies sampled on the same q-meshes in a given material from PhononDB and this work, respectively. This dimensionless metric quantifies the overall deviation in phonon frequencies between the two datasets for each material. The results are analyzed as a function of our calculated average phonon frequency $\bar{\omega} = \frac{1}{N} \sum_{i=1}^N \omega_i$, where N is the number of sampled modes. As shown in Fig. 2a, our calculated phonon frequencies agree well with those reported in PhononDB: more than 90% (705 out of 773) exhibit $\rho \leq 5\%$. However, a small number of outliers with ρ larger than 5% do exist. We highlight two representative cases: Be_2SrN_2 (Materials Project ID mp-11919) with $\rho = 5\%$, represents the majority of materials with good agreement, while ReO_3 (Materials Project ID mp-190), with a large deviation of $\rho = 32\%$, serves as an illustrative outlier. Figure 2b compares the phonon spectra for ReO_3 from both datasets. The primary differences lie in the acoustic region, where PhononDB reports several imaginary modes, whereas our results show fully real, positive frequencies. To confirm the source of this discrepancy, we repeat the phonon calculations using PhononDB's computational parameters—particularly its coarse k-mesh $2 \times 2 \times 2$ for supercell SCF calculations, which is significantly coarser than our k-mesh $4 \times 4 \times 4$ (automatically generated based on KSPACING = $0.15 \text{ \AA}^{-1}$). As shown in Figure S5, this reproduces the imaginary modes reported in PhononDB. Conversely, using the denser $4 \times 4 \times 4$ k-mesh with all other settings from PhononDB unchanged, we eliminate the imaginary modes. This confirms that the discrepancy stems from the underconverged k-mesh. A denser k-mesh in our workflow improves the accuracy of the internal force calculations, thereby reducing artificial imaginary phonon modes and improving the physical reliability of the phonon spectrum. Additionally, we note that PhononDB adopts a default plane-wave cutoff energy (ENCUT) of 1.3 times the largest ENMAX among constituent elements, which might be insufficient for certain materials and results in incomplete relaxation or force sampling errors. In contrast, we apply a consistent high ENCUT of 520 eV across all materials to ensure consistency and convergence. Nevertheless, for most materials, the computational settings used in PhononDB are sufficiently accurate, as evidenced by the agreement in Be_2SrN_2 (Fig. 2c). To provide a global view of the benchmark, we present a mode-by-mode frequency comparison in Fig. 2d. Our results demonstrate strong overall consistency with PhononDB, with $R^2 = 0.999$ and mean absolute error (MAE) of 0.13 THz. Most of the discrepancies are concentrated in the low-frequency regime, including imaginary modes, as exemplified by the ReO_3 case.

With the reliability of phonon frequency results confirmed, we next look into the lattice thermal conductivity (κ_L) calculated by our workflow.

Fig. 2 | Benchmark of phonon frequencies and lattice thermal conductivity against existing databases and experiments. **a** Mode-averaged phonon-frequency deviations between this work and PhononDB. Examples of compounds with **b** significant (ReO_3) and **c** moderate (Be_2SrN_2) frequency differences compared to PhononDB. **d** Mode-resolved comparison of phonon frequencies across two datasets. **e** Benchmark of calculated κ_L against reported experimental and DFT values from the literature. **f** Variation in calculated κ_L using different levels of lattice dynamics theory. Materials are ordered by increasing $\kappa_L^{\text{HA}+3\text{ph}}$. For visual clarity, bars for NaBH_4 with $R_{\text{HA}+3\text{ph}}^{\text{SCPH}+3\text{ph}} = 8.7$, $R_{\text{HA}+3\text{ph}}^{\text{SCPH}+3,4\text{ph}} = 7.0$, and $R_{\text{HA}+3\text{ph}}^{\text{SCPH}+3,4\text{ph}+\text{OD}} = 8.6$ are truncated at 3.0.



To construct a benchmark dataset, we select 33 representative materials from a diverse combination of the number of atoms in the unit cell and space groups. This benchmark dataset is also used in the convergence test of our workflow parameters (see details in Supplementary Information). Through an extensive literature review, we identify 25 of these 33 materials with previously reported κ_L values either calculated by DFT approaches^{45–52} or measured experimentally^{53–69}. Our calculated values are plotted against these reference values in Fig. 2e (detailed value comparisons in Table S6). It is worth noting that DFT-calculated κ_L could vary significantly due to the implementation of different levels of theory (this will be discussed thoroughly in later sections). When compared to DFT-calculated literature values, we align the theoretical level of our calculations with that of the literature whenever possible to ensure a physically meaningful comparison. When compared to the experimentally measured values, we use the highest level of theory in our work, which is $\kappa_L^{\text{SCPH}+3,4\text{ph}+\text{OD}}$. As shown in Fig. 2e, we find that all our calculated values fall within a reasonable range between 50% and 200% of the reported values. As expected, our results tend to overestimate κ_L relative to experiment, which can be attributed to real-world sample imperfections such as impurity scattering, grain boundaries, porosity, and finite-size effects that are currently absent in our computational framework. Given the high-throughput nature of our study, the observed level of agreement is deemed acceptable for screening and analysis purposes. Although our calculated materials show a diverse chemistries and structures, our use of a

consistent, well-converged computational protocol ensures that all results are self-consistent and directly comparable. This uniformity enables promising qualitative and quantitative findings based on the hierarchy of lattice dynamics theories within our dataset.

Within the framework of our consistent high-throughput calculations, a central question is how different levels of lattice dynamics theory affect the computed lattice thermal conductivity. In some cases, inclusion of anharmonic effects beyond the harmonic approximation yields significantly different values, whereas in others, even $\kappa_L^{\text{SCPH}+3,4\text{ph}+\text{OD}}$ has negligible difference from $\kappa_L^{\text{HA}+3\text{ph}}$. To systematically investigate these effects, we define a set of hierarchical ratios that quantify the impact of successive theoretical refinements, including HA + 3ph, SCPH + 3ph, SCPH + 3, 4ph, and SCPH + 3, 4ph + OD thermal conductivity:

$$R_{\text{HA}+3\text{ph}}^{\text{SCPH}+3\text{ph}} = \frac{\kappa_L^{\text{SCPH}+3\text{ph}}}{\kappa_L^{\text{HA}+3\text{ph}}}, \quad (3)$$

$$R_{\text{HA}+3\text{ph}}^{\text{SCPH}+3,4\text{ph}} = \frac{\kappa_L^{\text{SCPH}+3,4\text{ph}}}{\kappa_L^{\text{HA}+3\text{ph}}}, \quad (4)$$

$$R_{\text{HA}+3\text{ph}}^{\text{SCPH}+3,4\text{ph}+\text{OD}} = \frac{\kappa_L^{\text{SCPH}+3,4\text{ph}+\text{OD}}}{\kappa_L^{\text{HA}+3\text{ph}}} = \frac{\kappa_L^{\text{SCPH}+3,4\text{ph}} + \kappa_L^{\text{OD}}}{\kappa_L^{\text{HA}+3\text{ph}}}. \quad (5)$$

As demonstrated in Fig. 2f, these ratios show strong material dependence. In many cases, for example, FeNbSb, higher-order anharmonic effects have relatively modest impact: the calculated $\kappa_L^{\text{HA}+3\text{ph}}$ for FeNbSb is $26.12 \text{ W m}^{-1} \text{ K}^{-1}$, which is already close to the experimental value of $17.68 \text{ W m}^{-1} \text{ K}^{-1}$. In stark contrast, materials like NaBH₄ exhibit dramatic deviations. Its $\kappa_L^{\text{HA}+3\text{ph}}$ is only $0.20 \text{ W m}^{-1} \text{ K}^{-1}$, whereas the experimental κ_L is approximately $1.00 \text{ W m}^{-1} \text{ K}^{-1}$ ¹⁵⁷. Upon incorporating quartic anharmonicity, the calculated value increases to $\kappa_L^{\text{SCPH}+3,4\text{ph}} = 1.36 \text{ W m}^{-1} \text{ K}^{-1}$, yielding a ratio $R_{\text{HA}+3\text{ph}}^{\text{SCPH}+3,4\text{ph}} = 7.0$. This striking improvement clearly demonstrates the critical role of quartic anharmonicity in accurately capturing thermal transport in highly anharmonic systems. There are even more extreme cases where materials, such as anti-perovskite Ca₃NP, tetragonal perovskite SrHfO₃, oxide PbO, and noble-gas halide KrF₂, have imaginary phonon modes in harmonic phonon dispersions so that their $\kappa_L^{\text{HA}+3\text{ph}}$ are unavailable in Fig. 2f. Only by including the SCPH corrections at 300 K, the imaginary modes get hardened and make the κ_L calculations beyond HA level feasible for comparisons in Fig. 2e. Other materials exhibit less dramatic, but still meaningful, anharmonic effects, such as Ag₂TlI₂ ($R_{\text{HA}+3\text{ph}}^{\text{SCPH}+3,4\text{ph}+\text{OD}} = 2.0$), PbTiO₃ ($R_{\text{HA}+3\text{ph}}^{\text{SCPH}+3,4\text{ph}+\text{OD}} = 1.3$), and Cu₃VSe₄ ($R_{\text{HA}+3\text{ph}}^{\text{SCPH}+3,4\text{ph}+\text{OD}} = 0.8$), where including strong quartic anharmonicity also improves agreement with experimental values^{55,65,66}. Interestingly, there are also cases like Cs₂SnBr₆, where SCPH and four-phonon scattering, though both significant, cancel out each other, makes $R_{\text{HA}+3\text{ph}}^{\text{SCPH}+3,4\text{ph}+\text{OD}}$ close to unity. In general, when benchmark materials are ordered by increasing $\kappa_L^{\text{HA}+3\text{ph}}$ as shown in Fig. 2f, a clear trend emerges: the lower the $\kappa_L^{\text{HA}+3\text{ph}}$, the larger the deviation between theoretical levels. This implies that SCPH and four-phonon effects are more pronounced in low- κ_L materials—to some extent consistent with the intuitive notion that stronger lattice anharmonicity is typically associated with lower κ_L . However, notable exceptions exist. For instance, ZnTe, ZnS, and BeSe all exhibit anomalously strong four-phonon scattering effects that substantially suppress κ_L , even though their calculated κ_L are relatively high. Prior studies^{70–72} attribute this behavior to unusual phonon dispersion features, such as a wide acoustic-optical phonon gap, acoustic phonon bunching, and flat optical phonon bands, which suppress three-phonon processes and potentially enhance the relative contribution of four-phonon scattering. We indeed observe these features in the above three materials (see Fig. S6) and will discuss this theory thoroughly in a later section.

Our benchmark results emphasize more than just improved accuracy with higher-level theories. In fact, our results suggest a sophisticated influence from higher-order anharmonicity that is hard to conclude for all the materials. While such complexities can be more explicitly analyzed in a case-by-case study, the situation becomes considerably more intricate in a high-throughput setting, where materials span broad structural and chemical diversity. In this context, the selection of an appropriate theoretical level is not trivial. Although incorporating all higher-order anharmonic effects (SCPH, four-phonon scattering, and off-diagonal contributions) can enhance numerical agreement with experiment, these methods are computationally expensive, particularly for low-symmetry systems. There is thus a pressing need to evaluate in a broader context:

- To what extent do higher-order anharmonic effects impact κ_L ?
- Are such advanced levels of theory always necessary to achieve meaningful numerical accuracy?
- What physical insights can be extracted from systematically exploring this hierarchy of anharmonic effects?

Insights into the hierarchy of $\kappa_L^{\text{HA}+3\text{ph}}$, $\kappa_L^{\text{SCPH}+3\text{ph}}$, and $\kappa_L^{\text{SCPH}+3,4\text{ph}}$
To address the three core questions above, we must untangle the complex impacts on κ_L from different levels of theory across our dataset. Specifically, we assess the trends associated with each theoretical level, then interpret these findings with representative compounds' chemical

and structural features. We begin by quantifying the deviation between HA phonon frequencies (ω_λ) and those renormalized by SCPH theory (Ω_λ) within the 562 materials dataset. We define their mean absolute percentage difference as:

$$\text{MAPD}_{\text{HA}}^{\text{SCPH}} = \frac{1}{N} \sum_{\lambda=1}^N \left| \frac{\Omega_\lambda - \omega_\lambda}{\omega_\lambda} \right| \times 100\%, \quad (6)$$

where N is the number of sampled phonon modes corresponding in both SCPH and HA phonon dispersions. We plot $\text{MAPD}_{\text{HA}}^{\text{SCPH}}$ with respect to averaged HA phonon frequency $\bar{\omega} = \frac{1}{N} \sum \omega_\lambda$ in Fig. 3a. Overall, most compounds show $\text{MAPD}_{\text{HA}}^{\text{SCPH}}$ lower than 5%, indicating moderate frequency shifts. However, more significant renormalization effects are observed in systems with low $\bar{\omega}$, suggesting that soft-bonded anharmonic atomic networks are more susceptible to strong temperature-induced frequency shifts. We highlight top 3 high- $\text{MAPD}_{\text{HA}}^{\text{SCPH}}$ compounds across different $\bar{\omega}$ regimes to give more diverse examples of how SCPH influences the phonon spectrum, which are B1-CsCl (28%), CsClO₄ (21%), and NaBH₄ (20%). The comparison between their harmonic and SCPH-corrected phonon dispersions (see Fig. S6a–S6c) reveals substantial renormalization across the spectrum. These compounds either undergo phase transitions or tend to decompose^{73–75}, reflecting an extreme manifestation of their soft bonding nature—structural instability.

Beyond the magnitude of frequency shifts, we also examine their directionality, i.e., phonon hardening versus softening. This can be further quantified by positive and negative mean percentage difference, respectively, defined as

$$\delta_\lambda = \frac{\Omega_\lambda - \omega_\lambda}{\omega_\lambda}, \quad (7)$$

$$\text{MPD}^+ = \frac{1}{N^+} \sum_{\lambda \delta_\lambda > 0} \delta_\lambda, \quad (8)$$

$$\text{MPD}^- = -\frac{1}{N^-} \sum_{\lambda \delta_\lambda < 0} \delta_\lambda, \quad (9)$$

where N^+ and N^- correspond to the number of sampled phonon modes showing positive and negative frequency shifts, respectively. Note that in our 562-materials dataset, the HA phonon frequencies ω_λ are always positive, thereby preventing unphysical comparisons. The results of MPD^+ and MPD^- for every material in the dataset are depicted in Fig. 3b. The majority of materials fall above the line $y = x$, indicating that phonon hardening dominates over softening in most cases. However, there are anomalous cases. For example, B1-CsCl, discussed earlier, exhibits dominating phonon softening. We also identify compounds showing exclusively phonon hardening (t-AgI, Tl₂LiGaF₆, and KAgSe) or softening (ZnTe, InP, and AlSb). Detailed comparisons of their HA and SCPH phonon dispersions, along with a series of supporting analyses, are provided in the Supplementary Information.

While phonon spectra provide essential insights into lattice dynamics, they represent only one facet of the complex landscape governing phonon transport. To explicitly evaluate the influence of different levels of theory on κ_L , we perform a statistical analysis of the distribution of $R_{\text{HA}+3\text{ph}}^{\text{SCPH}+3\text{ph}}$ and $R_{\text{SCPH}+3\text{ph}}^{\text{SCPH}+3,4\text{ph}}$ across our 562-materials dataset:

$$R_{\text{SCPH}+3\text{ph}}^{\text{SCPH}+3,4\text{ph}} = \frac{\kappa_L^{\text{SCPH}+3,4\text{ph}}}{\kappa_L^{\text{SCPH}+3\text{ph}}} \quad (10)$$

These ratios independently quantify the most probable changes in κ_L induced by SCPH (Fig. 3c) and four-phonon scattering (Fig. 3d),

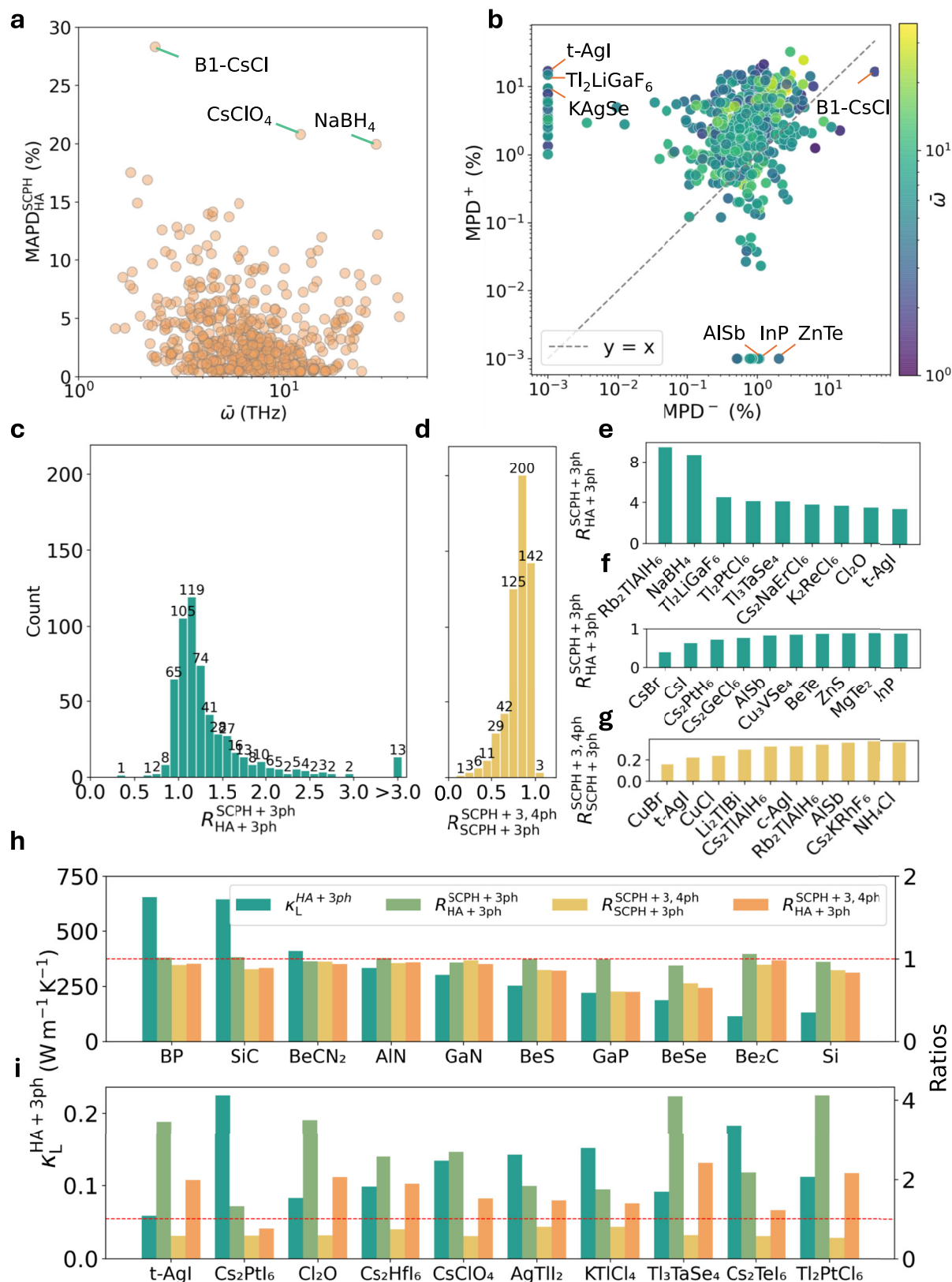


Fig. 3 | Impact of self-consistent phonon renormalization and four-phonon scattering on phonon frequencies and lattice thermal conductivity. a Mean absolute percentage difference (MAPD) between SCPH and HA phonon frequencies plotted against the average HA frequency for each compound. **b** Compound-wise comparison of positive and negative mean percentage differences (MPD⁺ and MPD⁻) in mode frequencies between SCPH and HA. **c** Distribution of $R_{\text{SCPH}+3\text{ph}}^{\text{HA}+3\text{ph}}$.

d Distribution of $R_{\text{SCPH}+3,4\text{ph}}^{\text{SCPH}+3\text{ph}}$. **e** Top-10 compounds ranked by highest $R_{\text{SCPH}+3\text{ph}}^{\text{SCPH}+3\text{ph}}$, **f** lowest $R_{\text{SCPH}+3\text{ph}}^{\text{SCPH}+3\text{ph}}$, and **g** lowest $R_{\text{SCPH}+3,4\text{ph}}^{\text{SCPH}+3\text{ph}}$. Comparison of $\kappa_L^{\text{HA}+3\text{ph}}$ and associated ratios for the **h** top-10 and **i** bottom-10 compounds ranked by $\kappa_L^{\text{SCPH}+3,4\text{ph}}$. Red dashed lines denote when ratio equals 1.

respectively. The off-diagonal heat transport contribution is discussed separately in the next section, as its impact is generally secondary compared to these two dominant effects. Figure 3c shows the distribution of $R_{\text{HA}+3\text{ph}}^{\text{SCPH}+3\text{ph}}$. Approximately 30% of compounds fall within the range of [0.9, 1.1], indicating relatively small changes in κ_L after SCPH correction. A majority, 68% exhibit $R_{\text{HA}+3\text{ph}}^{\text{SCPH}+3\text{ph}}$ larger than 1.1, suggesting a systematic increase in κ_L due to phonon hardening. Only 2% of compounds show $R_{\text{HA}+3\text{ph}}^{\text{SCPH}+3\text{ph}}$ lower than 0.9, corresponding to phonon softening and κ_L reduction. These findings are consistent with the results in Fig. 3b, where most phonon modes shift to higher frequencies after SCPH corrections, leading to increased group velocity, reduced scattering, and thus enhanced κ_L . We highlight the top-10 high- $R_{\text{HA}+3\text{ph}}^{\text{SCPH}+3\text{ph}}$ compounds in Fig. 3e. These include compounds from diverse chemistries—halides, selenides, hydrides, and oxides. Some of them are reported with metastability (t-AgI)⁷⁶, or share structural motifs with famous strong-anharmonicity systems, such as Ti_3TaSe_4 , which is isostructural with Ti_3VSe_4 ⁷⁷. Two extreme cases, hydride $\text{Rb}_2\text{TiAlH}_6$ and NaBH_4 , show over 8-fold increase in κ_L after SCPH correction, which is likely driven by highly ionic bonding environments. These cases are discussed in further detail in a later section. Meanwhile, we identify 12 compounds with an anomalous $R_{\text{HA}+3\text{ph}}^{\text{SCPH}+3\text{ph}} \leq 0.9$. These compounds unconventionally show phonon softening upon SCPH correction. Ten of these are listed in Figure 3f. They include reported unstable rocksalts (B1-CsBr and B1-CsI)⁷³, hydride and halide double-perovskites (Cs_2PtH_6 and Cs_2GeCl_6), and III-V group zinc blends (AlSb and InP). Interestingly, perovskites with similar structures can show opposite SCPH effects depending on their chemistry. For example, Ti_2PtCl_6 exhibits a 4-fold increase in κ_L after SCPH corrections, while Cs_2PtH_6 and Cs_2GeCl_6 show 29% and 24% reduction, respectively. This contrast highlights the need for in-depth SCPH analyses incorporating more advanced phonon renormalization theories, such as the second-order “bubble” diagram and non-diagonal self-energy corrections²². For the four-phonon scattering contribution, the distribution shown in Figure 3d reveals a clear trend: all compounds exhibit $R_{\text{SCPH}+3\text{ph}}^{\text{SCPH}+3,4\text{ph}} \leq 1$, confirming the physical expectation that four-phonon interactions always increase scattering and reduce κ_L . The top-10 low- $R_{\text{SCPH}+3\text{ph}}^{\text{SCPH}+3,4\text{ph}}$ compounds are shown in Figure 3g. These are predominantly halides (CuBr, t-AgI, c-AgI, CuCl) and double perovskites ($\text{Cs}_2\text{TiAlH}_6$, $\text{Rb}_2\text{TiAlH}_6$, and Cs_2KRhF_6)—systems with relatively low-frequency phonon modes and soft bonding environments that promote strong anharmonicity.

To illustrate the combined influence of SCPH and four-phonon scattering effects, we highlight the top-10 and bottom-10 compounds ranked by $\kappa_L^{\text{SCPH}+3,4\text{ph}}$. In the top-10 high κ_L compounds (Figure 3h, in $\kappa_L^{\text{SCPH}+3,4\text{ph}}$ descending order), we observe several well-known systems with strong bonding characteristics, such as III–V group compounds (BP, AlN, GaN, and GaP), silicon and silicon carbide, and Be-containing compounds (BeCN_2 , BeS, BeSe, and Be_2C). As expected, most of these materials exhibit $R_{\text{HA}+3\text{ph}}^{\text{SCPH}+3\text{ph}}$, $R_{\text{SCPH}+3\text{ph}}^{\text{SCPH}+3,4\text{ph}}$, and $R_{\text{HA}+3\text{ph}}^{\text{SCPH}+3,4\text{ph}}$ close to unity, indicating negligible influence from higher-order anharmonicity. For such systems, it is computationally efficient and justifiable to rely only on HA + 3ph level of theory for estimating κ_L without significant loss of fidelity. However, exceptions to this trend reveal important subtleties. GaP and BeSe show substantial reductions in κ_L upon inclusion of four-phonon scattering. The understanding of the origin of this phenomenon evolves progressively. It was first proposed by Lindsay et al.⁷⁰ that weak three-phonon interactions in BeSe stem from scattering channel limitation induced by two key phonon features: i) a wide acoustic-optical (a-o) phonon gap and ii) acoustic phonon bunching. Our phonon dispersion plots for GaP and BeSe in Figure S7 confirm the presence of both features. In our prior work²⁴, we investigated a system with similar features, HgTe, by artificially tuning its a-o gap. We found that changes in the a-o gap did not correlate with four-phonon scattering strength, suggesting that while a

wide a-o gap suppresses three-phonon scattering, it does not inherently promote four-phonon processes. Later, Xie et al.⁷² proposed that acoustic phonon bunching, specifically flat acoustic branches, can enhance resonant four-phonon scattering. This highlights acoustic bunching as a plausible mechanism behind the unexpectedly strong quartic effects in otherwise strongly bonded materials. Readers interested in more discussion about anomalously strong four-phonon scattering in high- κ_L compounds, including a correlation between $R_{\text{SCPH}+3\text{ph}}^{\text{SCPH}+3,4\text{ph}}$ and $\kappa_L^{\text{HA}+3\text{ph}}$, can refer to the Supplementary Information.

For the bottom-10 low- κ_L compounds (Fig. 3i, in $\kappa_L^{\text{SCPH}+3,4\text{ph}}$ ascending order), the higher-order anharmonicity plays a critical role in shaping the thermal transport. Take Ti_3TaSe_4 as an example, SCPH correction increases κ_L by a factor of four ($R_{\text{HA}+3\text{ph}}^{\text{SCPH}+3\text{ph}} = 4.1$), while four-phonon scattering reduces it approximately by half ($R_{\text{SCPH}+3\text{ph}}^{\text{SCPH}+3,4\text{ph}} = 0.59$). The resulting net effect is a doubling of κ_L ($R_{\text{HA}+3\text{ph}}^{\text{SCPH}+3,4\text{ph}} = 2.4$), indicating that neglecting higher-order anharmonicity would significantly underestimate κ_L and mis-rank compounds in high-throughput screening. Interestingly, these competing effects can also cancel out, leading to minimal net change in κ_L . For instance, in Cs_2TeI_6 , SCPH correction doubles κ_L ($R_{\text{HA}+3\text{ph}}^{\text{SCPH}+3\text{ph}} = 2.2$), while four-phonon scattering halves it ($R_{\text{SCPH}+3\text{ph}}^{\text{SCPH}+3,4\text{ph}} = 0.56$), resulting in a net impact ratio $R_{\text{HA}+3\text{ph}}^{\text{SCPH}+3,4\text{ph}} \approx 1$. In such cases, whether to accept the HA + 3ph result becomes a nuanced decision. It hinges on whether the high-throughput study is designed for efficient numerical screening of extreme κ_L values, or for extracting physical insight into complex anharmonic mechanisms. It is also worth noting that this work is limited up to quartic anharmonicity due to computational constraints. For systems exhibiting strong quartic anharmonicity, it is plausible that even higher-order interactions, such as fifth- and sixth-order anharmonicity, may become relevant.

Following the global statistical analysis of higher-order anharmonic effects, we now turn to three representative materials that exemplify extreme behaviors observed in the dataset. These are: (i) $\text{Rb}_2\text{TiAlH}_6$, which exhibits an exceptionally large SCPH-driven increase in κ_L with $R_{\text{HA}+3\text{ph}}^{\text{SCPH}+3\text{ph}} = 9.46$; (ii) Cu_3VSe_4 , which shows anomalous SCPH-driven suppression of κ_L with $R_{\text{HA}+3\text{ph}}^{\text{SCPH}+3\text{ph}} = 0.84$; (iii) CuBr, where κ_L is significantly reduced by four-phonon scattering with $R_{\text{SCPH}+3\text{ph}}^{\text{SCPH}+3,4\text{ph}} = 0.16$. We analyze each case in terms of crystal structure, phonon dispersion, phonon density of states (PDOS), phonon linewidth, cumulative κ_L , three-phonon (P3) or four-phonon (P4) phase space, and SCPH interaction parameter $I_{\lambda\lambda_1}$, as shown in Fig. 4. This comprehensive view allows us to pinpoint which phonon modes and corresponding atomic vibrations are responsible for the observed anomalies.

$\text{Rb}_2\text{TiAlH}_6$ (Fig. 4a) adopts the classic elpasolite (double-perovskite) structure (space group Fm3m, No. 225), in which Ti^{4+} ions occupy the 4a sites and Al^{3+} the 4b sites, each octahedrally coordinated to six hydride ions (H^-) located on 24e positions, while Rb^+ cations reside in the eightfold-coordinated 8c cavities. The bonding in $\text{Rb}_2\text{TiAlH}_6$ is predominantly ionic, comprising Rb^+-H^- , $\text{Ti}^{4+}-\text{H}^-$, and $\text{Al}^{3+}-\text{H}^-$ electrostatic interactions, with the AlH_6 and TiH_6 octahedra forming a three-dimensional corner-sharing network. As shown in Fig. 4b, the SCPH-corrected phonon dispersion at 300 K exhibits pronounced hardening across the entire spectrum compared to the 0 K harmonic one. This behavior aligns with previously reported PtH_8 , which also shows highly ionic and polarized metal-hydrogen bonding environment. The PDOS reveals that acoustic and low-frequency optical modes (below 5 THz) are primarily associated with the localized vibrations of rattler-like cations (Rb^+ and Ti^{4+}), whereas high-frequency optical modes (above 5 THz) are dominated by hydrogen. The frequency shift is particularly prominent for hydrogen-related modes, likely because hydrogen is the lightest element thus its thermally excited vibrations are far more significant than other elements. Meanwhile, we note that the extent of phonon hardening could be somewhat overestimated or underestimated owing to the diagonal-only approximation we made in SCPH. Despite much larger frequency shifts in high-frequency modes, the cumulative- κ_L curves (Fig. 4c) indicate that phonon modes below 5 THz contribute over 90% of the total thermal conductivity. Thus, the large increase in κ_L after

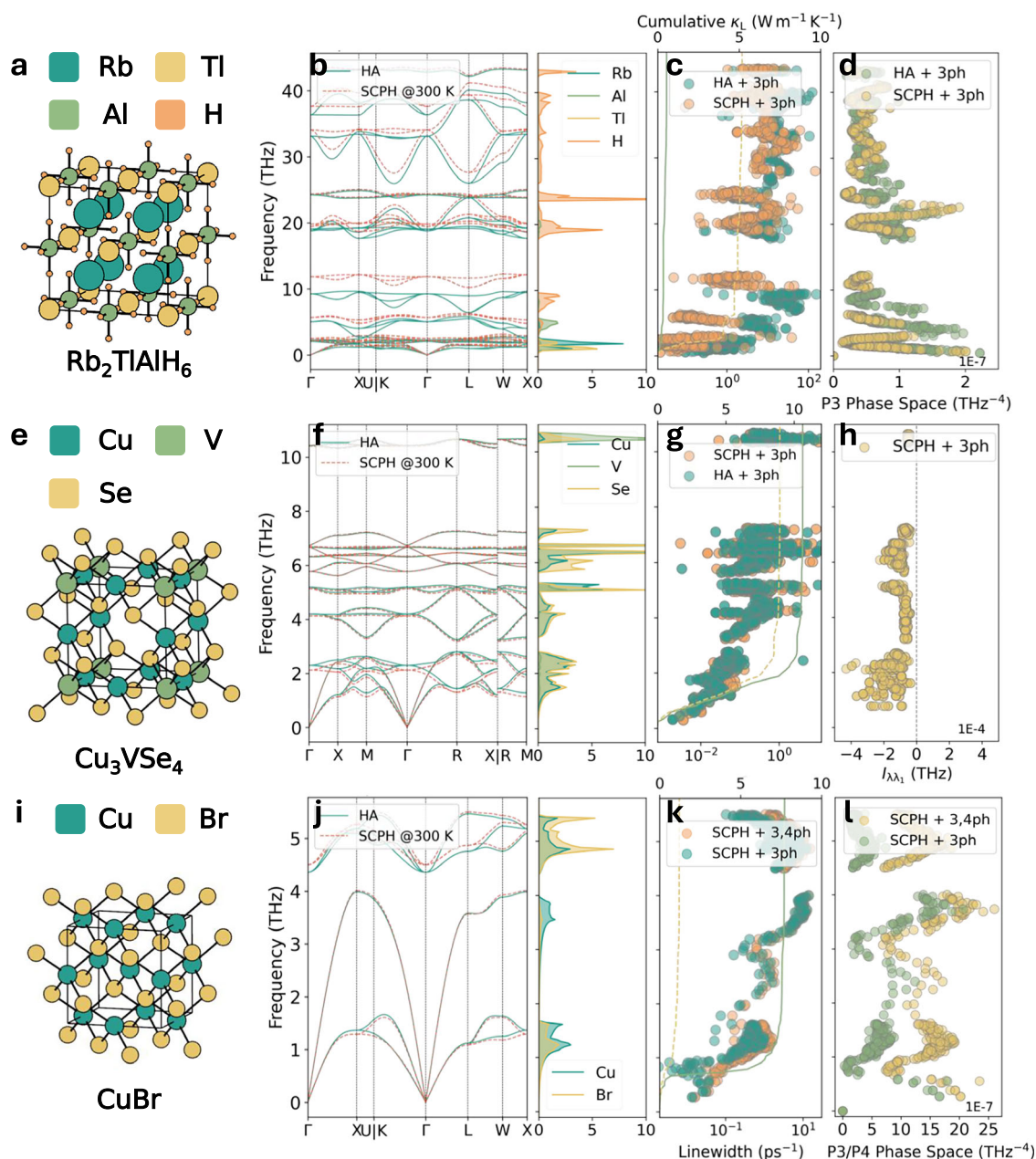


Fig. 4 | Microscopic analysis of lattice dynamics and thermal transport in representative materials with strong anharmonicity. Crystal structures, phonon dispersion, and frequency-dependent phonon density of states, linewidth and cumulative κ_L (HA + 3ph: green solid lines; SCPH+3(4)ph: yellow dashed lines),

and P3/P4 phase space or SCPH interaction parameter $I_{\lambda\lambda_1}$ for **a–d** $\text{Rb}_2\text{TlAlH}_6$ ($R_{\text{HA}+3\text{ph}}^{\text{SCPH}+3\text{ph}} = 9.46$), **e–h** Cu_3VSe_4 ($R_{\text{HA}+3\text{ph}}^{\text{SCPH}+3\text{ph}} = 0.84$), and **i–l** CuBr ($R_{\text{SCPH}+3,4\text{ph}}^{\text{SCPH}+3\text{ph}} = 0.16$).

SCPH correction is primarily due to the hardening of the cationic low-frequency modes. As shown in Fig. S8a, a comparison of mode-resolved group velocities before and after SCPH correction reveals that while phonon frequencies shift significantly, the dispersion shapes, i.e. the group velocities, of major heat-carrying acoustic modes remain largely unchanged. Therefore, the increase of κ_L after SCPH correction is mainly attributed to the decrease of low-frequency phonon linewidths, which originates from a decreased P3 scattering phase space (Fig. 4d) tightened by overall phonon frequency hardening. The case of $\text{Rb}_2\text{TlAlH}_6$ highlights that great SCPH hardening effect could come from the highly polarized ionic bonding. This effect might get further amplified by the presence of weak-bonding rattler-like ions, Rb^+ and Tl^+ . Although H^- vibration might not directly contribute to κ_L , it can still impact κ_L by shaping the scattering phase space—a collective property determined by the full phonon spectrum.

Cu_3VSe_4 , shown in Figure 4e, crystallizes in the cubic sylvanite structure (space group P43m, No. 215), in which V^{5+} resides at the 1a site and is tetrahedrally coordinated by four Se^{2-} ions (Wyckoff 4e), while Cu^+ occupies the 3d positions, each also tetrahedrally bound to Se^{2-} . The structure comprises corner-sharing VSe_4 and CuSe_4 tetrahedra, yielding a three-dimensional covalent/ionic network. Phonon dispersion comparisons in Fig. 4f reveal unusual phonon mode softening after SCPH correction at 300 K, which leads to anomalous $R_{\text{HA}+3\text{ph}}^{\text{SCPH}+3\text{ph}} = 0.84$. This behavior stands in sharp contrast to its well-studied analogue, Tl_3VSe_4 , which exhibits typical SCPH hardening with $R_{\text{HA}+3\text{ph}}^{\text{SCPH}+3\text{ph}} = 2^{38,77}$. Despite the similar stoichiometries, Cu_3VSe_4 and Tl_3VSe_4 (space group I43m, No. 217) have a major difference in crystal structures (see Fig. S9). Compared to Tl_3VSe_4 , where V^{5+} and Tl^+ reside at body center and face centers of the cubic unit cell, respectively, Cu_3VSe_4 leaves these sites unoccupied. The cumulative- κ_L

curves (Fig. 4g) tell that 90% of κ_L in Cu_3VSe_4 is contributed by phonons with frequencies below 2 THz, which, according to PDOS, are dominated by Cu and Se vibrations. Given the negligible change in group velocity (see Fig. S8b), the abnormal decrease in κ_L after SCPH correction must result from increased phonon linewidths due to the softening of acoustic and low-lying optical modes. To probe the origin of this unusual SCPH-induced softening, we focus on the most significantly softened transverse optical (TO) mode at the Brillouin zone center and analyze its interaction parameter $I_{\lambda\lambda_1}$ with all other phonon modes (Fig. 4h). Surprisingly, we find that $I_{\lambda\lambda_1}$ is negative for every mode λ_1 , indicating that all four-body interactions contribute to softening the TO mode. Based on Eq. (19), in the absence of imaginary modes in Cu_3VSe_4 phonon spectrum before and after SCPH renormalization, the renormalized frequencies Ω_λ and Ω_{λ_1} are always positive. Therefore, the uniformly negative values of $I_{\lambda\lambda_1}$ must stem from negative quartic IFCs $V^{(4)}$, meaning that the fourth-order interactions consistently lower the potential energy when atoms are displaced along the TO mode coordinate—directly contributing to phonon softening. Structurally, we have noted that Cu_3VSe_4 differs from its counterpart Ti_3VSe_4 by lacking both body-center and face-center atoms in the unit cell. These structural vacancies may create additional degrees of vibrational freedom, enabling strong quartic softening via SCPH corrections.

In CuBr, we observe remarkably strong four-phonon scattering, leading to a significantly suppressed thermal conductivity with $R_{\text{SCPH}+3\text{ph}}^{\text{SCPH}+3,4\text{ph}} = 0.16$. As depicted in Fig. 4i, CuBr crystallizes in the zinc blende (sphalerite) structure (space group $\bar{F}43m$, No. 216), where Cu^+ cations occupy the tetrahedral 2c sites and are coordinated to four Br^- anions at the 2a and equivalent positions. This arrangement yields a three-dimensional network of corner-sharing CuBr_4 tetrahedra, characterized by predominantly ionic Cu^+-Br^- interactions. Our previous study⁷⁹ has shown that $p-d$ orbital hybridization between Cu and Br leads to antibonding states below the Fermi level, weakening the Cu–Br bonds. This results in a soft lattice, which can be confirmed by phonon dispersions in Fig. 4j, where most of the phonon branches lie below 5 THz. While the existence of low-lying and flat phonon modes in CuBr has been previously discussed, the extremely high four-phonon scattering in this material has not been reported. It is worth noting that SCPH correction brings κ_L from $6.24 \text{ W m}^{-1} \text{ K}^{-1}$ (HA + 3ph) to $7.71 \text{ W m}^{-1} \text{ K}^{-1}$ (SCPH + 3ph), whereas four-phonon scattering drastically reduces κ_L to $1.22 \text{ W m}^{-1} \text{ K}^{-1}$ (SCPH + 3, 4ph), aligning closely with the experimental value of $1.25 \text{ W m}^{-1} \text{ K}^{-1}$ ¹⁸⁰. As shown in the phonon linewidth plot (Fig. 4k), total linewidth nearly doubles after four-phonon scattering is included (note the x-axis is on a log scale), indicating that the four-phonon contribution is comparable in magnitude to three-phonon scattering. Such high scattering rate stems from a large four-phonon (P4) phase space as suggested by P3/P4 phase space comparison in Fig. 4l. This behavior mirrors aforementioned findings in compounds BeSe ⁷⁰, HgTe ²⁴, and AgCrSe_2 ⁷², where the acoustic phonon bunching feature is linked to strong resonant four-phonon scattering. A similar bunching feature is evident in CuBr phonon dispersion around 1 THz, suggesting that resonant four-phonon scattering mechanisms may be at play here as well.

Off-diagonal thermal conductivity

We discuss κ_L^{OD} in a separate section because it differs fundamentally from the three levels of diagonal κ_L discussed previously. According to Eq. (21), κ_L^{OD} can be computed at any level of lattice dynamics theory, provided the necessary harmonic phonon properties—phonon populations, frequencies, group velocities—and the phonon linewidths are available. As we mentioned before, we consider κ_L^{OD} only at the SCPH + 3, 4ph level to simplify the analysis. To quantify the importance of κ_L^{OD} relative to $\kappa_L^{\text{SCPH}+3,4\text{ph}}$, we define the ratio:

$$R_{\text{SCPH}+3,4\text{ph}}^{\text{OD}} = \frac{\kappa_L^{\text{OD}}}{\kappa_L^{\text{SCPH}+3,4\text{ph}}} \quad (11)$$

As shown in Fig. 5a, out of the 562 materials in our dataset, 414 exhibit a κ_L^{OD} contribution amounting to less than 10% of the corresponding $\kappa_L^{\text{SCPH}+3,4\text{ph}}$. This suggests that for the majority of materials, κ_L^{OD} is negligible and may be safely ignored in practical calculations. To identify when off-diagonal contributions become non-negligible, we plot $R_{\text{SCPH}+3,4\text{ph}}^{\text{OD}}$ as a function of $\kappa_L^{\text{SCPH}+3,4\text{ph}}$ on a log scale in Fig. 5b. The ratio exhibits a clear inverse relationship with $\kappa_L^{\text{SCPH}+3,4\text{ph}}$: as the diagonal thermal conductivity increases beyond $10 \text{ W m}^{-1} \text{ K}^{-1}$, $R_{\text{SCPH}+3,4\text{ph}}^{\text{OD}}$ rapidly decays toward zero. This trend highlights that off-diagonal thermal conductivity is mostly significant in low- κ_L materials, which typically exhibit stronger anharmonicity and broader phonon linewidths. Among the high $R_{\text{SCPH}+3,4\text{ph}}^{\text{OD}}$ compounds, we present the top 10 with the largest off-diagonal contributions in Fig. 5c. Cl_2O stands out with the highest $R_{\text{SCPH}+3,4\text{ph}}^{\text{OD}}$ of 1.43, making it the only case in our dataset where $\kappa_L^{\text{SCPH}+3,4\text{ph}+\text{OD}}$ exceeds $\kappa_L^{\text{SCPH}+3,4\text{ph}}$. However, Cl_2O is known to be unstable under ambient conditions and undergoes photolytic decomposition into Cl_2 and O_2 at a moderate rate⁸¹, making its room-temperature thermal transport analysis less convincing. We instead focus on a secondary representative, KTiCl_4 ($R_{\text{SCPH}+3,4\text{ph}}^{\text{OD}} = 0.75$), which is stable at room temperature and pressure⁸². As depicted in Fig. 5d, KTiCl_4 crystallizes in a zircon-type structure (space group $I4_1/a$, No. 88), where Ti^{3+} ions occupy the 4a Wyckoff positions and K^+ resides on 4b sites, while Cl^- anions populate 16f positions. The Ti^{3+} ion is tetrahedrally coordinated by four Cl^- ions, forming distorted TiCl_4 units, whereas the K^+ ion forms a twisted square-antiprism KCl_6 unit with eight neighboring Cl^- ions (four shorter and four longer contacts). This highly distorted and ionic bonding network contributes to a strongly anharmonic lattice.

The exceptionally high $R_{\text{SCPH}+3,4\text{ph}}^{\text{OD}}$ in KTiCl_4 can be attributed to two primary factors. The first is the broad linewidths and low $\kappa_L^{\text{SCPH}+3,4\text{ph}}$. As illustrated by phonon dispersions (Fig. 5e) and the cumulative κ_L curve (Fig. 5f), more than 90% of heat is carried by phonons below 4 THz. PDOS suggests that acoustic phonons are dominated by Ti vibrations within a very narrow frequency window (0–1 THz), aligned with the well-known rattling behavior of Ti in prior reports^{77,83}. Meanwhile, the low-lying optical phonons are dominated by Cl and K vibrations, owing to their lighter masses compared to Ti. Moreover, these modes exhibit strong SCPH-induced frequency shifts, suggesting high lattice anharmonicity. As a result, the phonon linewidths are significantly broadened in Fig. 5f, and the calculated $\kappa_L^{\text{SCPH}+3,4\text{ph}}$ is only $0.21 \text{ W m}^{-1} \text{ K}^{-1}$, providing a small denominator for $R_{\text{SCPH}+3,4\text{ph}}^{\text{OD}}$ and amplifying the relative contribution of κ_L^{OD} .

The second reason is the highly overlapping distribution of phonon mode less than 4 THz. The phonon spectrum reveals a dense cluster of modes below 4 THz with minimal dispersion. This quasi-degenerate mode distribution facilitates strong coupling between closely spaced phonon modes. In such a regime, wave-like tunneling of vibrational energy becomes efficient, especially when the mode energy differences are small and linewidths are broad. This is further validated in Fig. 5g, where we plot the $\omega_s - \omega_{s'}$ pairwise contributions to κ_L^{OD} . Dominant contributors cluster near the diagonal line, indicating strong coupling between quasi-degenerate phonon pairs with $\omega_s \approx \omega_{s'}$. We emphasize that this plot excludes diagonal elements ($s \neq s'$), so any high contributions with $\omega_s = \omega_{s'}$ arise from distinct but degenerate phonon modes providing viable tunneling channels, rather than conventional particle-like transport. For comparison, we plot the pairwise contribution to diagonal ($s = s'$) thermal conductivity $\kappa_L^{\text{SCPH}+3,4\text{ph}}$ in Fig. 5h, which suggests that moderate diagonal thermal conductivity come from acoustic phonons below 1 THz, further confirming that thermal transport in KTiCl_4 is dominated by the wave-like tunneling of highly overlapping, low-lying optical phonon modes with broad linewidths.

Given the negligible computational cost of evaluating κ_L^{OD} , and despite its limited impact in most materials, we recommend computing it

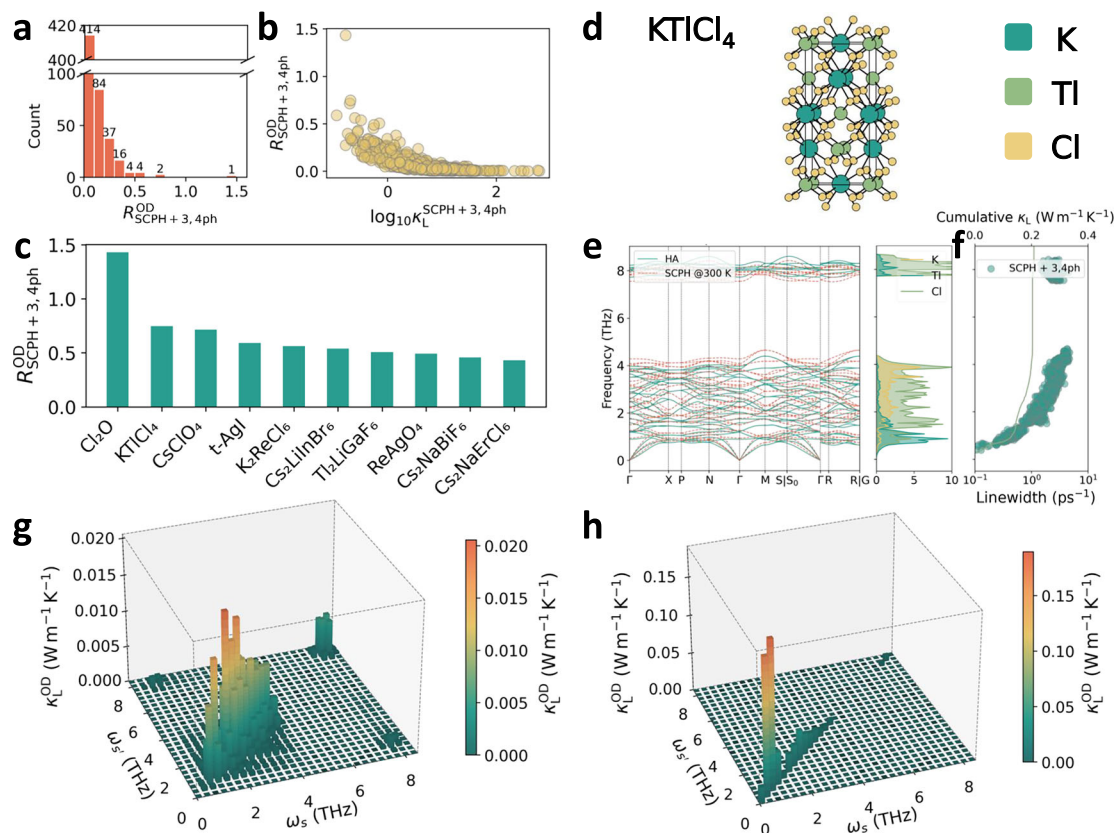


Fig. 5 | Significance and microscopic analysis of off-diagonal thermal transport. **a** Distribution of $R_{\text{SCPH}+3,4\text{ph}}^{\text{OD}}$. **b** Dependence of $R_{\text{SCPH}+3,4\text{ph}}^{\text{OD}}$ on the logarithm of $\kappa_{\text{L}}^{\text{SCPH}+3,4\text{ph}}$. **c** Top 10 compounds with the highest $R_{\text{SCPH}+3,4\text{ph}}^{\text{OD}}$. **d** Crystal structure of KTiCl_4 . Phonon transport properties of KTiCl_4 : **e** phonon dispersion at 0 K (HA)

and 300 K (SCPH) and density of states at 0 K (HA), **f** cumulative κ_{L} and phonon linewidths versus phonon frequency, **g, h** Spectral decomposition of the off-diagonal and diagonal contribution to κ_{L} as a function of phonon mode pair frequencies ω_s and $\omega_{s'}$. Frequency above 8 THz are truncated due to negligible contributions.

in all cases for theoretical completeness and consistency, as advocated in our previous work⁴².

Discussion

We have demonstrated a state-of-the-art high-throughput computational workflow that enables accurate and efficient first-principles prediction of hierarchical lattice dynamics and lattice thermal conductivities for large materials sets. This automated pipeline computes interatomic force constants up to the 4th order, incorporates self-consistent phonon renormalization and three-/four-phonon scattering processes, and solves the phonon Boltzmann transport equation with full diagonal and off-diagonal contributions.

A comparison between our calculated phonon frequencies and those from PhononDB suggests high consistency ($R^2 = 0.999$). A second benchmark with 33 literature-reported κ_{L} indicates our predictions at the SCPH + 3, 4ph + OD level are within a reasonable range. However, we emphasize again that the principal aim of this workflow is to generate a hierarchical phonon frequency and κ_{L} database that can be self-consistently compared in a uniform framework, instead of predicting the κ_{L} close to experimental measurement. We apply this workflow to a diverse set of crystalline compounds spanning a wide range of chemistries and structural prototypes. This diversity in chemistry (from elementals to quaternaries involving 73 elements) and structure (38 different cubic and tetragonal space groups) ensures that our conclusions are broadly applicable across material classes.

A key novelty of this work is the hierarchical lattice dynamics and thermal conductivity dataset we established for each material, systematically capturing the incremental impact of higher-order anharmonic effects. For every compound, we obtained harmonic and renormalized phonon spectra respectively at 0 K and 300 K, and a “family” of thermal

conductivity values: from the baseline harmonic approximation with three-phonon scattering ($\kappa_{\text{L}}^{\text{HA}+3\text{ph}}$), to including SCPH correction ($\kappa_{\text{L}}^{\text{SCPH}+3\text{ph}}$), to including four-phonon scattering ($\kappa_{\text{L}}^{\text{SCPH}+3,4\text{ph}}$), and finally the full solution with off-diagonal heat flux terms ($\kappa_{\text{L}}^{\text{SCPH}+3,4\text{ph}+\text{OD}}$). We introduce a series of metrics, including $\text{MAPD}_{\text{HA}}^{\text{SCPH}}$, MPD^+ and MPD^- , $R_{\text{HA}+3\text{ph}}^{\text{SCPH}+3\text{ph}}$, $R_{\text{SCPH}+3\text{ph}}^{\text{SCPH}+3,4\text{ph}}$, and $R_{\text{SCPH}+3,4\text{ph}}^{\text{OD}}$ to quantify the impact of these key higher-order anharmonic effects.

Analyzing this hierarchy yields several important insights. First, we find that for the majority of materials ($\sim 60\%$ of the dataset), the HA + 3ph level is sufficiently close (within $\sim \pm 20\%$) to the fully anharmonic level, i.e. SCPH + 3, 4ph + OD. This suggests that expensive higher-order calculations can be bypassed in most cases without significant loss of accuracy. Second, the SCPH renormalization generally increases κ_{L} (by 0–30% in 60% of compounds, due to dominant phonon frequency hardening). We use $\text{Rb}_2\text{TiAlH}_6$ as an extreme example ($R_{\text{HA}+3\text{ph}}^{\text{SCPH}+3\text{ph}} = 9.46$), showing an exceptionally high SCPH effect due to the highly polarized ionic bonding within the system. We also identified anomalous cases where including SCPH actually decreased κ_{L} due to phonon softening. In particular, compounds such as Bi-CsBr , Cs_2PtH_6 , and Cu_3VSe_4 exhibit this unusual behavior, with SCPH lowering their κ_{L} by roughly 10–60%. Third, as expected, four-phonon scattering always acts to reduce κ_{L} . The magnitude of the four-phonon reduction varies: for high- κ_{L} materials it is modest (often a 5–15% decrease), whereas in very anharmonic low- κ_{L} materials it can reach 30–84% suppression of κ_{L} . We use CuBr as an extreme case which shows $R_{\text{SCPH}+3\text{ph}}^{\text{SCPH}+3,4\text{ph}} = 0.16$, indicating the soft bonding-induced Cu–Br p – d hybridization brings significant quartic anharmonicity to the system. There

are also exceptions. We observe unusually strong four-phonon scattering rates in high- κ_L compounds ZnTe, ZnS, and BeSe with $R_{\text{SCPH}+3\text{ph}}^{\text{SCPH}+3,4\text{ph}}$ of 0.5, 0.55, and 0.6, respectively, which might be related to resonant four-phonon scattering induced by acoustic phonon bunching. Given that similar acoustic phonon bunching is also observed in low- κ_L compounds such as HgTe, AgCrSe₂, and CuBr, this could be a universal feature promising strong four-phonon scattering. Fourth, we confirm a strong correlation between off-diagonal contributions and the absolute thermal conductivity—wave-like tunneling transport is significant only in low- κ_L solids. In our dataset, materials with $\kappa_L^{\text{SCPH}+3,4\text{ph}+\text{OD}}$ below $1 \text{ W m}^{-1} \text{ K}^{-1}$ show substantial off-diagonal effects (accounting for up to 60% of total κ_L), whereas moderate- and high- κ_L crystals (κ_L above $\sim 10 \text{ W m}^{-1} \text{ K}^{-1}$) have negligible off-diagonal fractions < 5%. We select KTiCl₄ as an example ($R_{\text{SCPH}+3,4\text{ph}}^{\text{OD}} = 0.75$), demonstrating that a highly anharmonic system (broad phonon linewidths) with clustered phonon bands (small vibration energy difference) tends to facilitate the off-diagonal heat transport. All these insights help us tailor the strategy when performing high-throughput κ_L calculations for a certain species of materials.

Meanwhile, we want to note several limitations of our study: i) The major trends reported here are derived from high-symmetry systems (cubic and tetragonal), chosen due to limited computational resources. Extending to lower-symmetry crystals (hexagonal, orthorhombic, monoclinic, etc.) is expected to reveal richer quartic or even higher-order anharmonic effects, requiring more elaborate treatments of phonon renormalization, phonon-phonon scattering, and off-diagonal transport. ii) Our κ_L analysis considered dynamical stability but not thermodynamic stability under ambient conditions. For example, molecular crystals such as Cl₂O (mp-29537) may decompose at room temperature, making their reported κ_L values of limited physical meaning. While we have confirmed that such materials are of minority and do not impact the general trend we observed, users should carefully examine the finite-temperature stability before applying any specific compounds from our dataset. iii) To maintain a uniform baseline workflow applicable across broad chemical and structural space, we did not include system-specific settings such as van der Waals (vdW) corrections, special exchange-correlation functional and smearing scheme, non-analytical corrections (NAC), or spin polarization. Our tests (detailed in the Supplementary Information) indicate that neglecting some of these (e.g., NAC) generally introduces non-essential deviations of κ_L and does not alter the overall conclusions. Nevertheless, when applying our workflow to specific systems where such corrections are essential, we recommend re-evaluating with the corresponding corrections incorporated.

In summary, our work demonstrates a major step forward in predictive lattice thermal conductivity modeling, marrying comprehensiveness (accounting for phonon renormalization, three- and four-phonon scattering, and off-diagonal heat flux) with high-throughput efficiency. The resulting database of phonon frequencies and thermal conductivities, each with a quantified hierarchy of anharmonic effects, not only provides immediate insights into heat transport mechanisms, but also serves as a rich resource for data-driven studies. In the future, this dataset will open the door to machine-learning models that go beyond direct κ_L prediction: one could train models to predict the hierarchy metrics (such as the ratios $R_{\text{HA}+3\text{ph}}^{\text{SCPH}+3\text{ph}}$, $R_{\text{SCPH}+3,4\text{ph}}^{\text{SCPH}+3,4\text{ph}}$, and $R_{\text{HA}+3\text{ph}}^{\text{SCPH}+3,4\text{ph}}$ that quantify the influence of SCPH and four-phonon processes on κ_L). Such predictive capability would be transformative as an “anharmonic importance classifier”—allowing us to rapidly assess, given only a crystal structure, whether a simple HA + 3ph calculation will suffice or whether full SCPH + 3, 4ph + OD treatment is necessary for accurate thermal conductivity predictions. Ultimately, the combination of our high-throughput workflow and ML-guided screening can accelerate the discovery and thermal characterization of materials by focusing computational efforts where they matter most, thereby advancing both fundamental understanding and practical design of materials with desired heat transport properties.

Methods

Compressive-sensing force fitting

In a lattice at equilibrium, the force F acting on atom a can be expanded as a Taylor series with respect to atomic displacements. Truncating the expansion at the fourth order yields:

$$F_a = F_0 + \Phi_{ab}^{(2)} u_b + \frac{1}{2!} \Phi_{abc}^{(3)} u_b u_c + \frac{1}{3!} \Phi_{abcd}^{(4)} u_b u_c u_d, \quad (12)$$

where $u_a \equiv u_{a,\alpha}$ denotes the displacement of atom a along the Cartesian direction α , and $\Phi^{(n)}$ represents the n th-order Interatomic force constants (IFCs). This expression can be written more compactly as:

$$\mathbf{F} = [\mathbb{A}^{(2)} \mathbb{A}^{(3)} \mathbb{A}^{(4)}] [\Phi^{(2)} \Phi^{(3)} \Phi^{(4)}]^\top, \quad (13)$$

where each $\mathbb{A}^{(n)}$ contains combinations of atomic displacements appropriate for the corresponding IFC order. For example, the fourth-order matrix $\mathbb{A}^{(4)}$ takes the form

$$\mathbb{A}^{(4)} = -\frac{1}{3!} \begin{bmatrix} u_b^1 u_c^1 u_d^1 & \cdots \\ \cdots & \ddots \\ u_b^L u_c^L u_d^L & \cdots \end{bmatrix}, \quad (14)$$

with superscripts indicating different lattice points.

To determine the IFCs, we exploit the sparsity inherent in force constant tensors and formulate the problem as a classic compressive sensing framework using the LASSO operator⁸⁴:

$$\Phi^{\text{CS}} = \arg \min_{\Phi} \|\Phi\|_1 + \frac{\mu}{2} \|\mathbf{F} - \mathbb{A}\Phi\|_2^2. \quad (15)$$

Here, $\|\Phi\|_1$ is the ℓ_1 -norm of the IFCs. $\|\mathbf{F} - \mathbb{A}\Phi\|_2^2$ is the squared ℓ_2 -norm of the residual between the DFT-calculated forces $\mathbf{F}$ and the model-predicted forces $\mathbb{A}\Phi$. Regularization parameter μ balances the trade-off between sparsity (ℓ_1 -norm minimization) and accuracy (ℓ_2 -norm minimization). The $\arg \min$ minimization operator acts on the sum of ℓ_1 -norm and ℓ_2 -norm terms. In our previous study³⁸, we employ a “cocktail”-flavor fitting strategy to mitigate the “pre-renormalization” of lower-order IFCs: first, the second-order IFCs $\Phi^{(2)}$ are obtained using the finite-displacement method (FDM) with particularly small displacements; then the corresponding contribution $\mathbb{A}^{(2)}\Phi^{(2)}$ is subtracted from the total force $\mathbf{F}$; finally $\Phi^{(3)}$ and $\Phi^{(4)}$ are fit using the residual forces as in Eq. (16). This fitting scheme is also adopted in the present work.

T-dependent displacement generation

Once the harmonic $\Phi^{(2)}$ are available, the corresponding phonon frequencies ω_λ can be computed by diagonalizing the dynamical matrix. These frequencies then enable the generation of temperature-dependent atomic displacements $\mathbb{A}$ using the quantum covariance matrix Σ_{u_a, u_b} defined for atoms a and b in the original supercell as²¹:

$$\Sigma_{u_a, u_b} = \frac{\hbar}{2\sqrt{m_a m_b}} \sum_{\lambda} \frac{(1 + 2n_{\lambda}^0)}{\omega_{\lambda}} e_a^{\lambda} e_b^{\lambda*}, \quad (16)$$

where n_{λ}^0 is the Bose-Einstein distribution function, and λ indexes the phonon mode by wave vector $\mathbf{q}$ and branch s . Here, m_a and e_a^{λ} are the atomic mass and the phonon eigenvector projected onto atom a , respectively.

Self-consistent phonon renormalization

The renormalization of phonon frequencies ω_λ due to quartic anharmonicity at finite temperatures is computed using the self-consistent phonon renormalization (SCPH) theory⁸⁵. When formulated in reciprocal space, the first-order correction from quartic anharmonicity (retaining only diagonal

contributions) leads to the SCPH equation:

$$\Omega_\lambda^2 = \omega_\lambda^2 + 2\Omega_\lambda \sum_{\lambda_1} I_{\lambda\lambda_1}, \quad (17)$$

where Ω_λ is the renormalized frequency incorporating finite-temperature effects. The quantity $I_{\lambda\lambda_1}$ captures the anharmonic interaction between phonon modes λ and λ_1 , and is given by:

$$I_{\lambda\lambda_1} = \frac{\hbar}{8N} \frac{V^{(4)}(\lambda, -\lambda, \lambda_1, -\lambda_1)}{\Omega_\lambda \Omega_{\lambda_1}} \left[1 + 2n(\Omega_{\lambda_1}) \right], \quad (18)$$

where N , $\hbar$, and $V^{(4)}(\lambda, -\lambda, \lambda_1, -\lambda_1)$ are, respectively, the number of sampled wave vectors, the reduced Planck constant, and the reciprocal representation of the fourth-order interatomic force constants (see detailed expression in Eq. (29)). The temperature effects enter through the phonon population n , making the SCPH formalism temperature-dependent. Because Ω_λ appears on both sides of Eq. (18), and also in the denominator of $I_{\lambda\lambda_1}$, the renormalized frequencies must be determined self-consistently via iterative solution of the SCPH equation until convergence is achieved. Notably, we adopted the diagonal approximation in obtaining the SCPH solution, similar to the approach used in stochastic self-consistent harmonic approximation (SSCHA)⁸⁶.

Lattice thermal conductivity calculation

The lattice thermal conductivity κ_L can be calculated by solving the Peierls-Boltzmann transport equation (PBTE)²⁸ under the single-mode relaxation time approximation (SMRTA):

$$\kappa_L = \frac{1}{k_B T^2 \Lambda N} \sum_{\lambda} n_{\lambda}^0 (n_{\lambda}^0 + 1) (\hbar \omega_{\lambda})^2 \mathbf{v}_{\lambda} \otimes \mathbf{v}_{\lambda} \tau_{\lambda}, \quad (19)$$

where N , Λ and T are the number of sampled phonon wave vectors, the volume of the primitive cell, and the absolute temperature, respectively. $\mathbf{v}_{\lambda}$ and τ_{λ} denote the group velocity and phonon lifetime (the reciprocal of phonon linewidth/scattering rate).

Off-diagonal heat transport

To capture additional thermal transport contributions beyond the standard phonon gas model, particularly in materials with strong disorder or glass-like dynamics, we employ Simoncelli et al.'s formalism²³. In this approach, the wave-like phonon tunneling conduction arising from off-diagonal terms of the heat-flux operator can be captured by:

$$\kappa_L^{\text{OD}} = \frac{\hbar^2}{k_B T^2 \Lambda N} \sum_{\mathbf{q}} \sum_{s \neq s'} \frac{\omega_{\mathbf{q}}^s + \omega_{\mathbf{q}}^{s'}}{2} \mathbf{v}_{\mathbf{q}}^{s,s'} \otimes \mathbf{v}_{\mathbf{q}}^{s',s} \times \frac{\omega_{\mathbf{q}}^s n_{\mathbf{q}}^s (n_{\mathbf{q}}^s + 1) + \omega_{\mathbf{q}}^{s'} n_{\mathbf{q}}^{s'} (n_{\mathbf{q}}^{s'} + 1)}{4(\omega_{\mathbf{q}}^{s'} - \omega_{\mathbf{q}}^s)^2 + (\tau_{s,\mathbf{q}}^{-1} + \tau_{s',\mathbf{q}}^{-1})^2}, \quad (20)$$

where s and s' denote distinct phonon branches at the same wave vector $\mathbf{q}$. The quantity $\mathbf{v}_{\mathbf{q}}^{s,s'}$ is the generalized group velocity tensor, defined as:

$$\mathbf{v}_{\mathbf{q}}^{s,s'} = \frac{i}{\omega_{\mathbf{q}}^s + \omega_{\mathbf{q}}^{s'}} \sum_{\alpha, \beta} \sum_{m, p, q} e_{\mathbf{q}}^s(\alpha, p) D_{\beta\alpha}^{pq}(0, m) (\mathbf{R}_m + \mathbf{R}_{pq}) e^{iq \cdot \mathbf{R}_m} e_{\mathbf{q}}^{s'}(\beta, q), \quad (21)$$

where e , D , and $\mathbf{R}$ denote the polarization vector, the dynamical matrix, and the lattice vector, respectively. The indices α, β refer to Cartesian coordinates, m labels unit cell, and p, q index atoms within the unit cell.

Phonon lifetimes

With the temperature-dependent effective harmonic IFCs $\Phi^{(2)'} obtained from SCPH calculations, the phonon frequencies ω_{λ} (Ω_{λ}) and group velocities $\mathbf{v}_{\lambda}$ can be directly extracted. To evaluate the phonon lifetime τ_{λ} , we$

compute the total phonon scattering rates τ_{λ}^{-1} using Matthiessen's rule:

$$\tau_{\lambda}^{-1} = \tau_{\lambda, 3\text{ph}}^{-1} + \tau_{\lambda, 4\text{ph}}^{-1} + \tau_{\lambda, \text{iso}}^{-1}, \quad (22)$$

where $\tau_{\lambda, 3\text{ph}}^{-1}$, $\tau_{\lambda, 4\text{ph}}^{-1}$, and $\tau_{\lambda, \text{iso}}^{-1}$ represent the phonon scattering rates due to three-phonon interactions, four-phonon interactions, and isotope scattering respectively. Within SMRTA, these scattering rates can be expressed as⁴⁰:

$$\tau_{\lambda, 3\text{ph}}^{-1} = \sum_{\lambda_1 \lambda_2} \left[\frac{1}{2} (1 + n_{\lambda_1}^0 + n_{\lambda_2}^0) \zeta_+ + (n_{\lambda_1}^0 - n_{\lambda_2}^0) \zeta_- \right], \quad (23)$$

$$\tau_{\lambda, 4\text{ph}}^{-1} = \sum_{\lambda_1 \lambda_2 \lambda_3} \left[\frac{1}{6} \frac{n_{\lambda_1}^0 n_{\lambda_2}^0 n_{\lambda_3}^0}{n_{\lambda}^0} \zeta_{---} + \frac{1}{2} \frac{(1 + n_{\lambda_1}^0) n_{\lambda_2}^0 n_{\lambda_3}^0}{n_{\lambda}^0} \zeta_{+-} + \frac{1}{2} \frac{(1 + n_{\lambda_1}^0)(1 + n_{\lambda_2}^0) n_{\lambda_3}^0}{n_{\lambda}^0} \zeta_{++} \right], \quad (24)$$

where the three-phonon scattering matrix elements for absorption (+) and emission (−) processes can be further expressed as:

$$\zeta_{\pm} = \frac{\pi \hbar}{4N} |V^{(3)}(\lambda, \pm \lambda_1, -\lambda_2)|^2 \Delta_{\pm} \frac{\delta(\omega_{\lambda} \pm \omega_{\lambda_1} - \omega_{\lambda_2})}{\omega_{\lambda} \omega_{\lambda_1} \omega_{\lambda_2}}, \quad (25)$$

and the corresponding four-phonon scattering matrix elements for recombination (++), redistribution (+ −), and splitting (− −) processes are given by:

$$\zeta_{\pm \pm} = \frac{\pi \hbar^2}{8N^2} |V^{(4)}(\lambda, \pm \lambda_1, \pm \lambda_2, -\lambda_3)|^2 \Delta_{\pm \pm} \times \frac{\delta(\omega_{\lambda} \pm \omega_{\lambda_1} \pm \omega_{\lambda_2} - \omega_{\lambda_3})}{\omega_{\lambda} \omega_{\lambda_1} \omega_{\lambda_2} \omega_{\lambda_3}}, \quad (26)$$

where $V^{(3)}$ and $V^{(4)}$ are the third- and fourth-order IFCs expressed in reciprocal space as:

$$V^{(3)}(\lambda, \pm \lambda_1, -\lambda_2) = \sum_{a, p_1, b, p_2, c} \frac{e_a^{\lambda} e_b^{\pm \lambda_1} e_c^{-\lambda_2}}{\sqrt{m_a m_b m_c}} \Phi_{abc}^{(3)} e^{\pm i \mathbf{q}_1 \cdot \mathbf{r}_{p_1}} e^{-i \mathbf{q}_2 \cdot \mathbf{r}_{p_2}}, \quad (27)$$

and

$$V^{(4)}(\lambda, \pm \lambda_1, \pm \lambda_2, -\lambda_3) = \sum_{a, p_1, b, p_2, c, p_3, d} \frac{e_a^{\lambda} e_b^{\pm \lambda_1} e_c^{\pm \lambda_2} e_d^{-\lambda_3}}{\sqrt{m_a m_b m_c m_d}} \Phi_{abcd}^{(4)} e^{\pm i \mathbf{q}_1 \cdot \mathbf{r}_{p_1}} e^{\pm i \mathbf{q}_2 \cdot \mathbf{r}_{p_2}} e^{-i \mathbf{q}_3 \cdot \mathbf{r}_{p_3}}, \quad (28)$$

where p, m , and r index the primitive cell, atomic mass, and lattice vector, respectively. The momentum conservation conditions are enforced via δ functions: $\Delta_{\pm} \equiv \delta(\mathbf{q} \pm \mathbf{q}_1 - \mathbf{q}_2)$ and $\Delta_{\pm \pm} \equiv \delta(\mathbf{q} \pm \mathbf{q}_1 \pm \mathbf{q}_2 - \mathbf{q}_3)$. Energy conservation is implemented using a δ function approximated by adaptive Gaussian smearing⁸⁷ for three- and four-phonon processes. Recent methodological advancements⁴¹ further improve computational efficiency by reducing the number of required phonon scattering process samples. Specifically, irreducible quartets of phonon modes $\lambda, \lambda_1, \lambda_2$, and λ_3 can be statistically represented by a reduced, randomly sampled subset, enabling significant acceleration of solving Eqs. (25) and (27).

Data availability

Data availability: The 2nd-order interatomic force constants (IFCs) under harmonic approximation (HA) and after self-consistent phonon renormalization (SCPH) corrections; The 3rd- and 4th-order IFCs; mean absolute percentage errors (MAPE) of interatomic-force constant fitting; Phonon frequency mean absolute percentage difference (MAPD), positive percentage difference (MPD⁺), and negative percentage difference (MPD[−]); κ_L values at different levels of theory for 773 materials are available at <https://doi.org/10.5281/zenodo.15937696>.

Code availability

GitHub repositories for codes used in our workflow: [https://github.com/LLNL/cslD\(CSLD\)](https://github.com/LLNL/cslD(CSLD)); [https://phonopy.github.io/phonopy/\(Phonopy\)](https://phonopy.github.io/phonopy/(Phonopy)); [https://github.com/yimavxia/TD-Disp/tree/main\(TD-Disp\)](https://github.com/yimavxia/TD-Disp/tree/main(TD-Disp)); [https://github.com/FourPhonon\(FourPhonon\)](https://github.com/FourPhonon(FourPhonon)); [https://github.com/yimavxia/Unifiedkappa-phonopy\(UnifiedKappa\)](https://github.com/yimavxia/Unifiedkappa-phonopy(UnifiedKappa)).

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Author contributions

Y.X. and C.W. initiated the project and outlined the key components of the workflow. Z.L. and H.L. carried out the computational tasks, benchmarked parameter performance, and refined the workflow using optimized settings. Z.L. performed statistical analyses on the dataset. Y.X. provided the initial codes for the workflow. The research was overseen by Y.X. and C.W. All authors were involved in manuscript preparation, engaged in discussions, and contributed to revisions and feedback.

Competing interests

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Additional information

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